Amazon Data Firehose Hands-On Lab

To Begin with the Lab

Summary of the Lab

In this lab, **Amazon Data Firehose** was created to deliver streaming data into **Amazon S3**. The video explained that Firehose can either take data from **Amazon Kinesis Data Streams** or use **Direct PUT** as the source. For this practical lab, **Direct PUT** was selected so the stream could receive data directly without needing Kinesis Data Streams first. An S3 bucket was created as the destination, buffer settings were configured so Firehose would collect data for a short period before delivering it, and then the built-in **demo data producer** was used to test the flow. After the test ran, the streamed records appeared in the S3 bucket in folder-like prefixes, and the downloaded file could be opened to verify the data. The lab proved that Firehose can collect incoming data, buffer it, and write it to S3 automatically.

Understanding Amazon Data Firehose

Amazon Data Firehose is a fully managed service that receives streaming data and delivers it to destinations such as **Amazon S3**. It exists so you do not need to build your own streaming ingestion pipeline for simple delivery use cases. In this lab, Firehose acted as the middle service between the producer and S3. The producer sent data to Firehose, Firehose buffered the data, and then Firehose delivered it to S3 after the buffer interval or buffer size was reached.

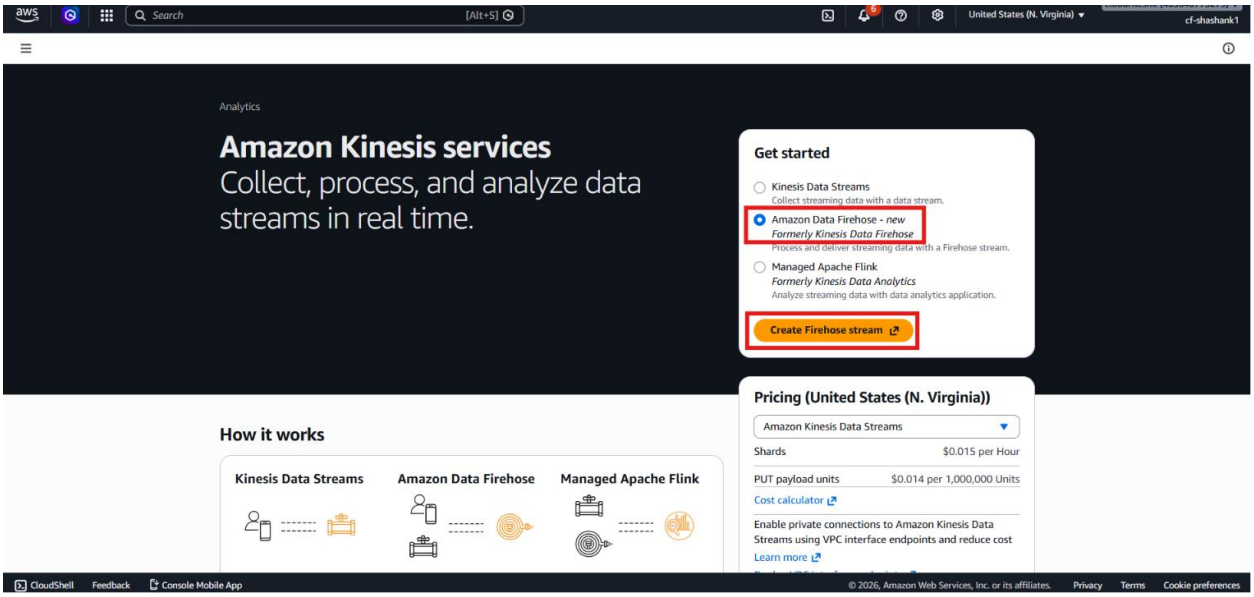
Example:

A stock ticker app sends small updates such as symbol, price, and timestamp. Firehose collects those updates for a few seconds and then writes them to an S3 bucket so the data can later be analyzed or archived.

Lab Steps

Step 1 – Open Amazon Data Firehose

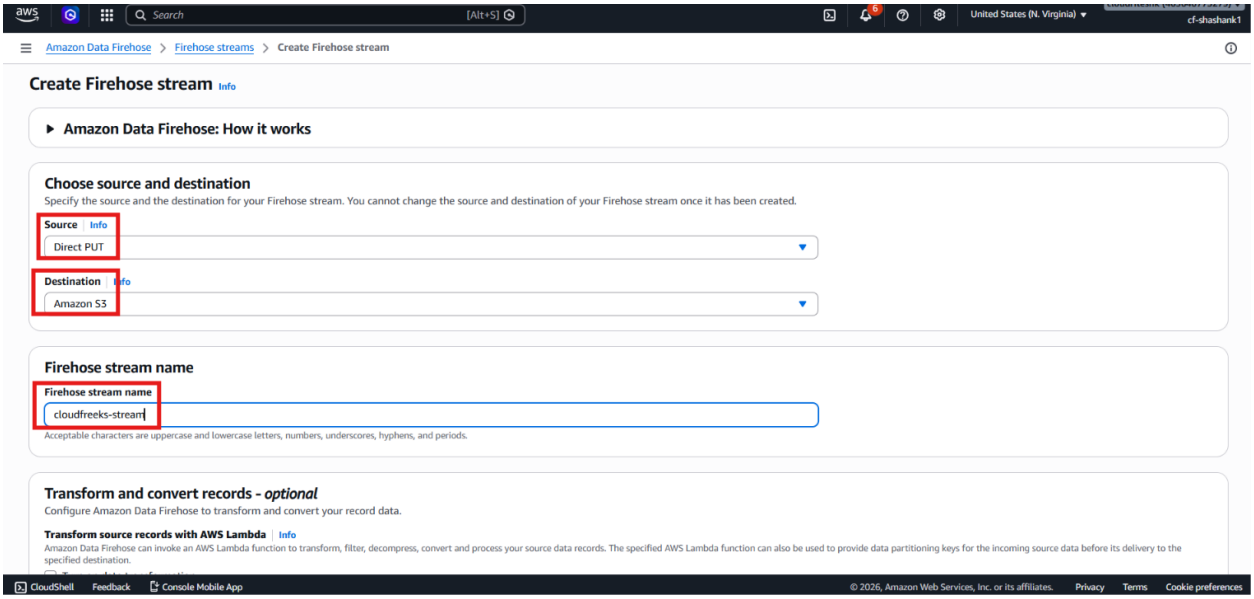
- Sign in to the AWS Management Console.
- Open **Amazon Kinesis**.
- Select **Amazon Data Firehose**.



- Click **Create Firehose stream**.

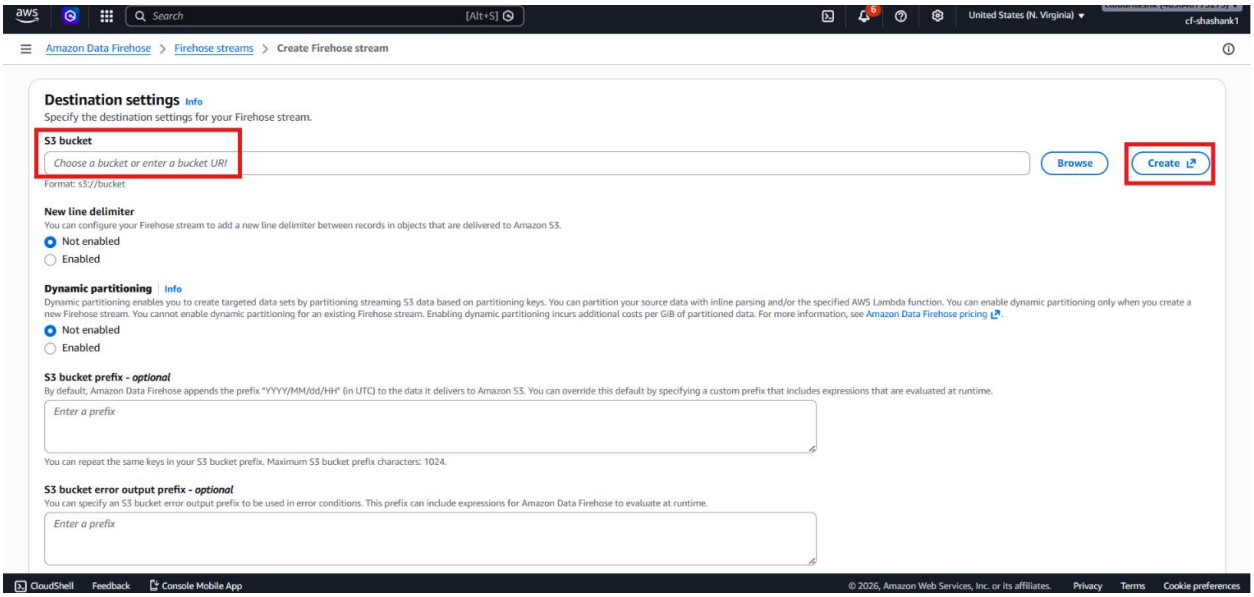
Step 2 – Choose the Source

- Select **Direct PUT** as the source.
- Understand that Firehose can also use **Kinesis Data Streams** as a source, but that was not used in this lab.
- Enter the Firehose stream name: cloudfreeks-stream

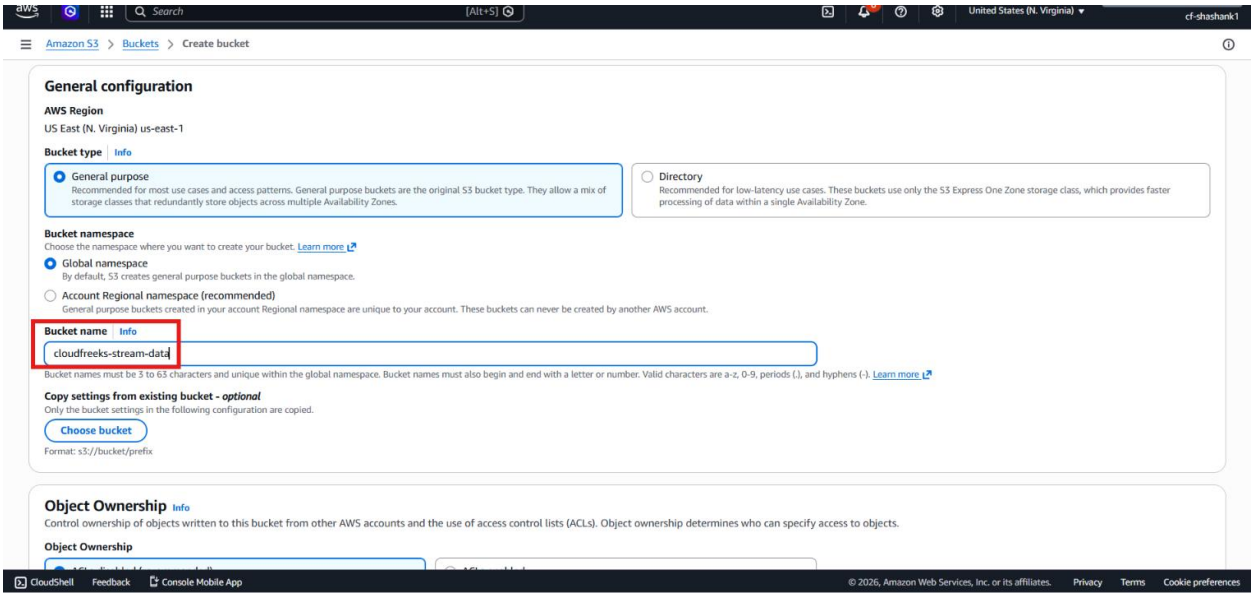


Step 3 – Choose the Destination

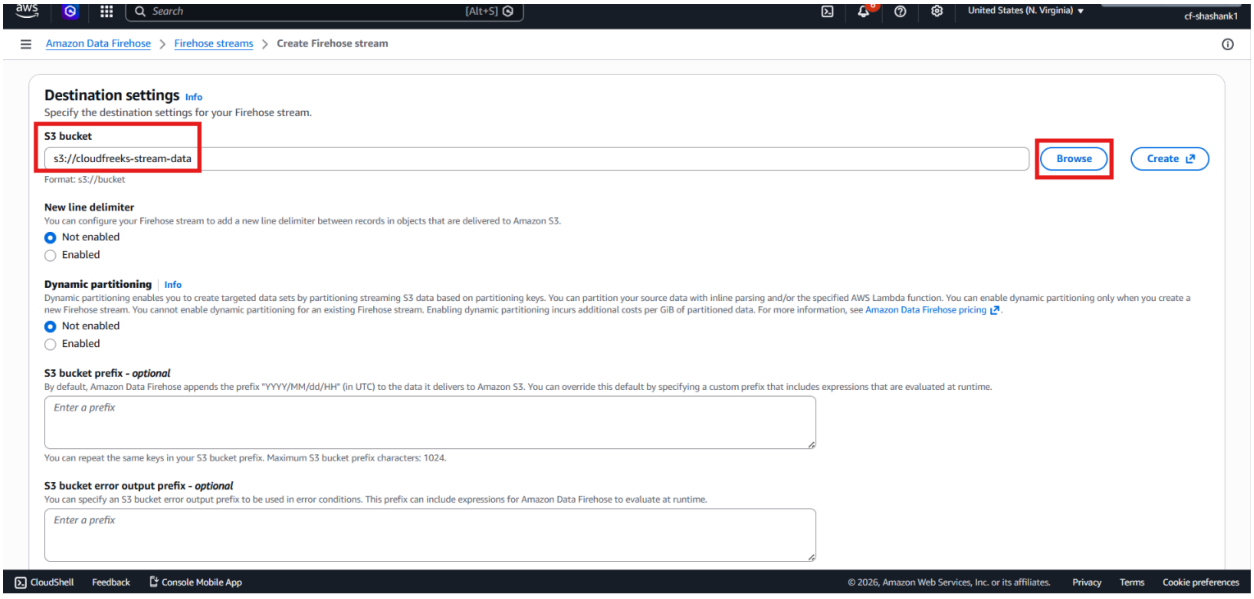
- Select **Amazon S3** as the destination.
- Create a new S3 bucket if needed.



- Enter the bucket name: cloudfreeks-stream-data

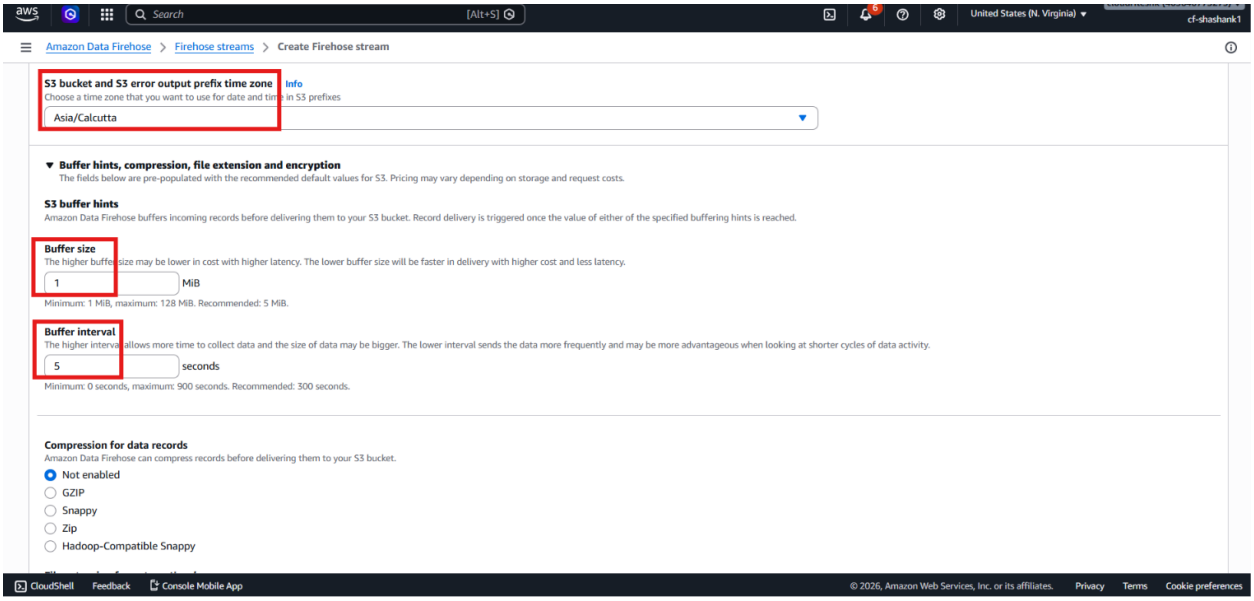


- Browse and select the bucket.
- Keep the destination configuration simple.

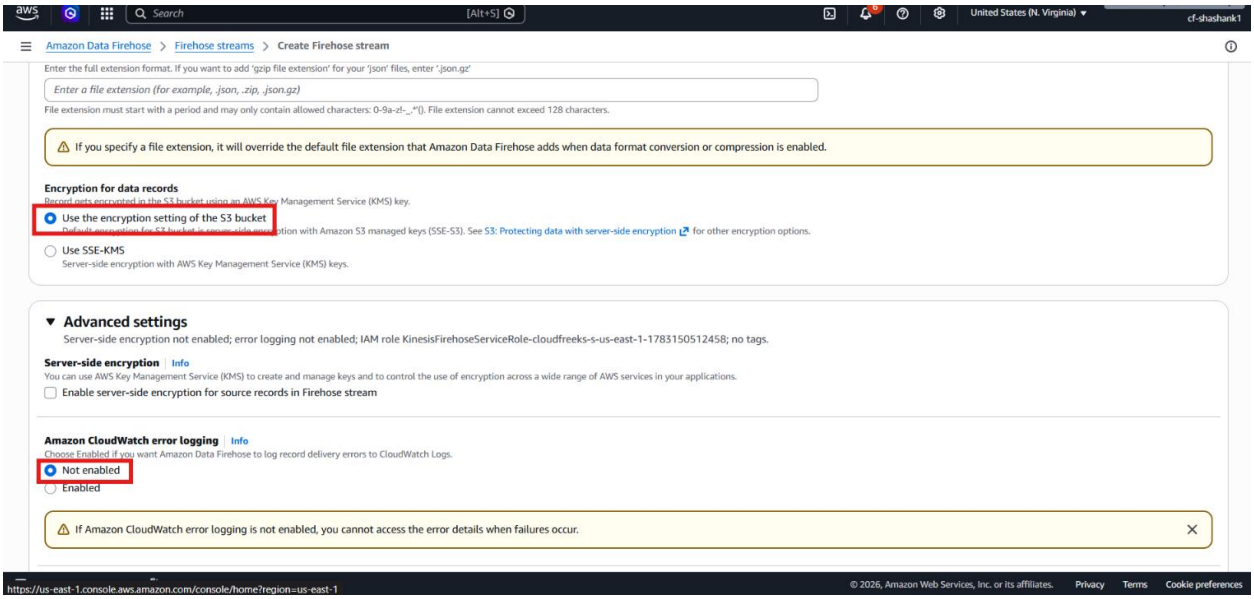


Step 4 – Configure Buffer Settings

- Leave transformation and conversion options disabled.
- Set the buffer size to **1 MiB**.
- Set the buffer interval to **5 seconds**.
- Choose the appropriate timezone, such as **Asia/Kolkata** for India.

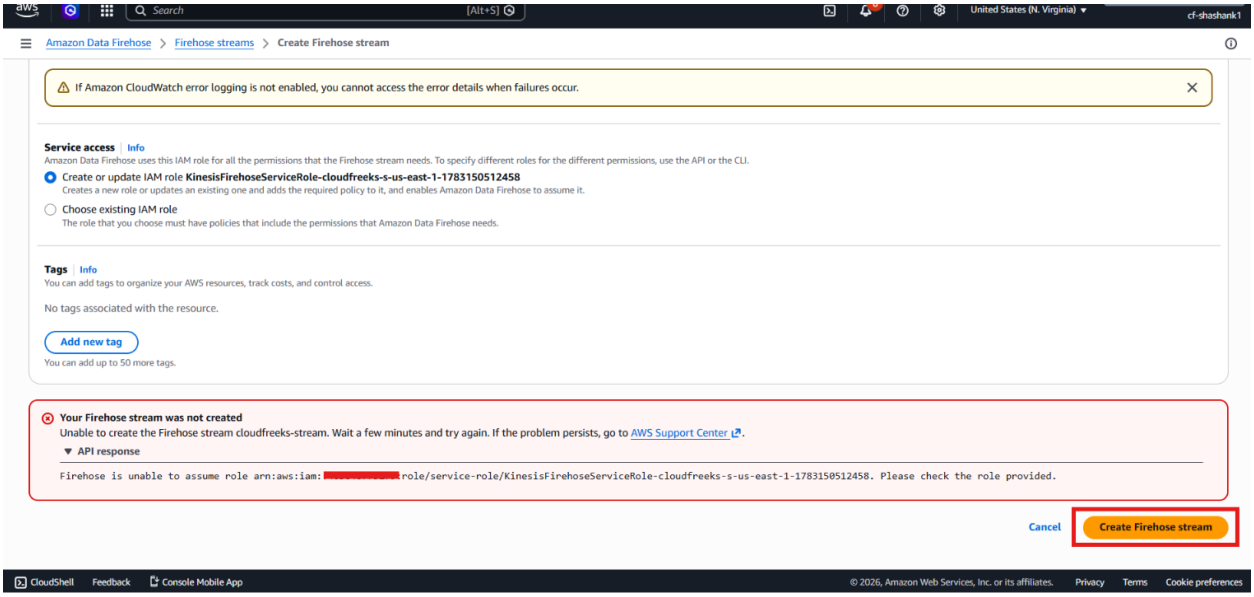


- Keep encryption as the default S3 encryption.
- Leave CloudWatch error logging disabled for this lab.

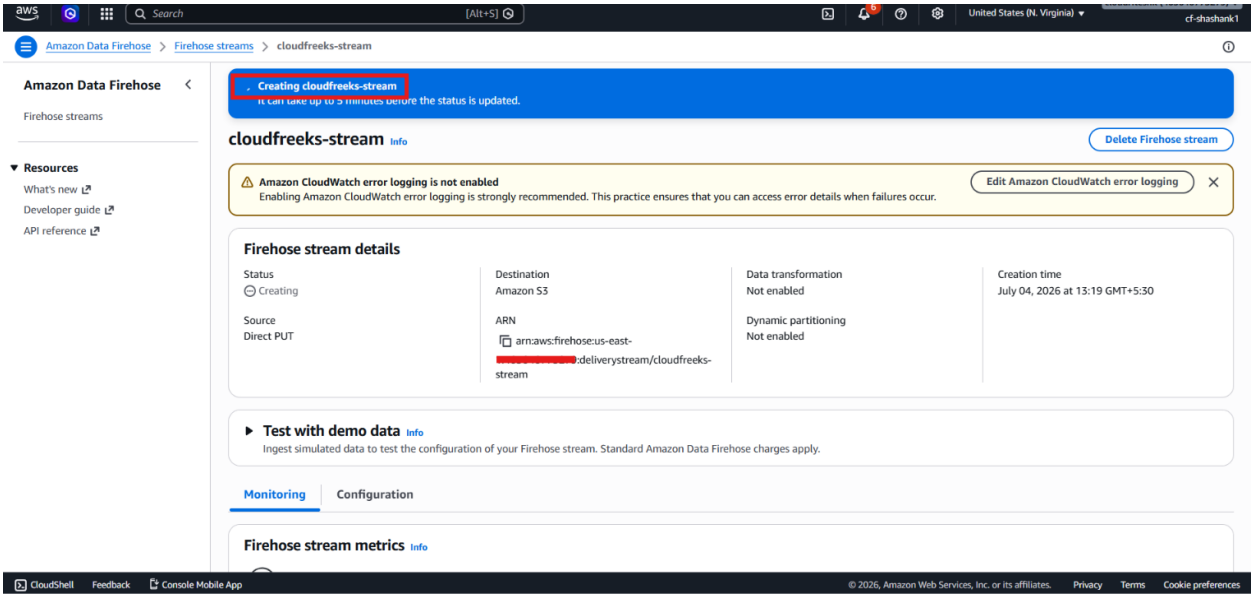


Step 5 – Create the Firehose Stream

- Click **Create Firehose stream**.
- If AWS shows an error the first time, click **Create Firehose stream** again.

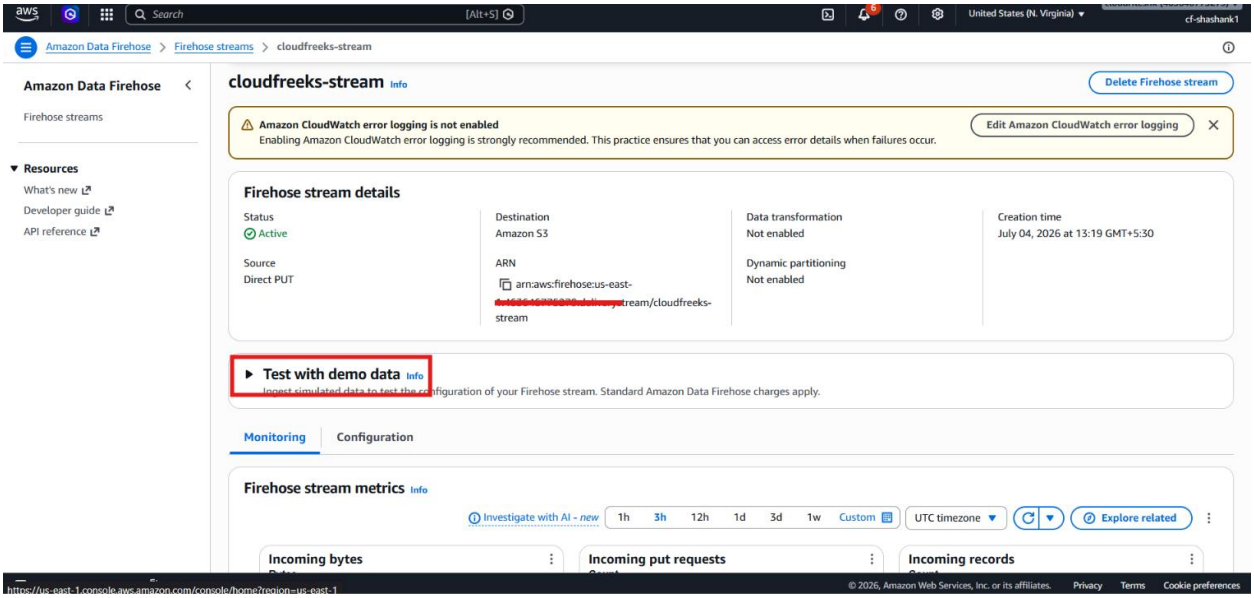


- Wait until the stream becomes **Active**.

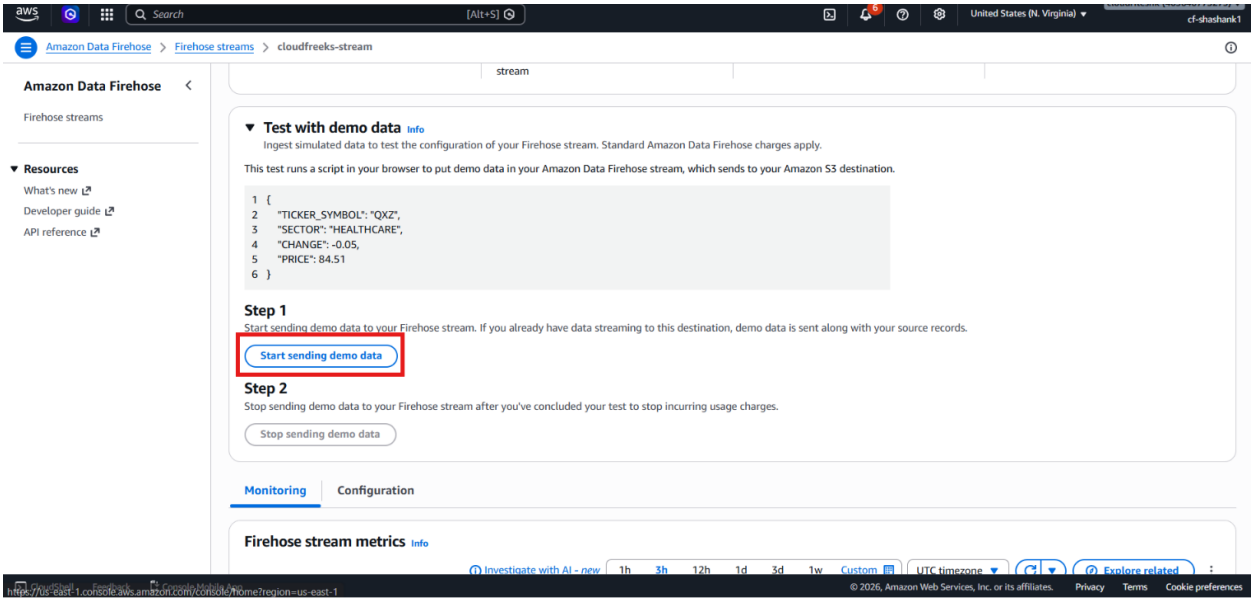


Step 6 – Test with Demo Data

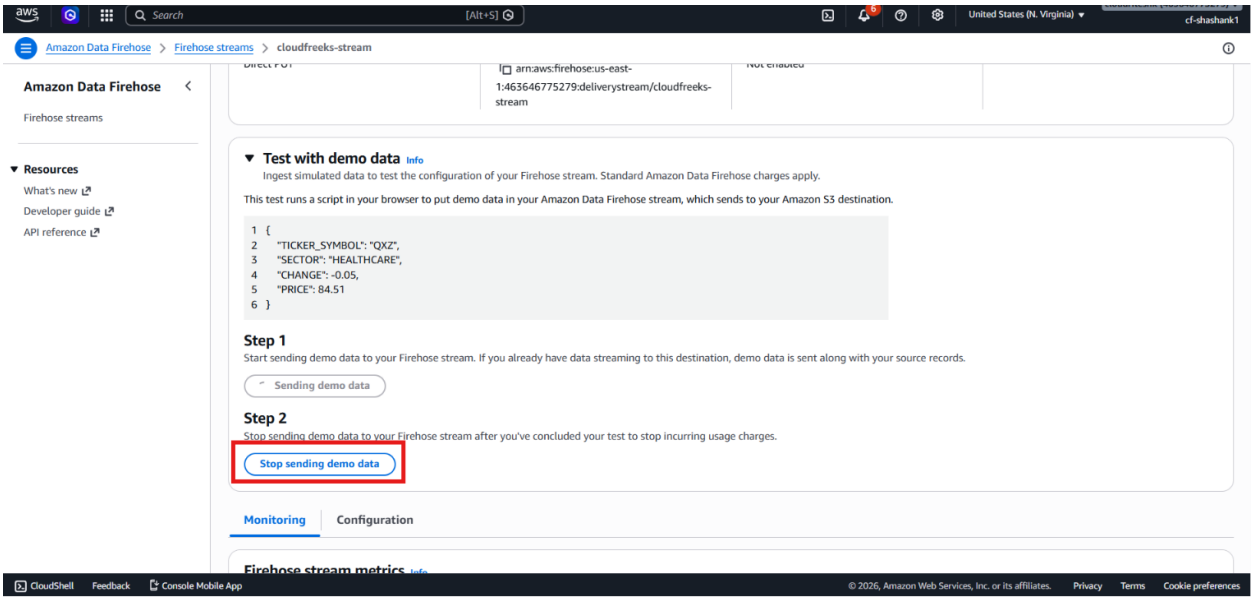
- Open **Test with demo data**.



- Click **Start sending demo data**.



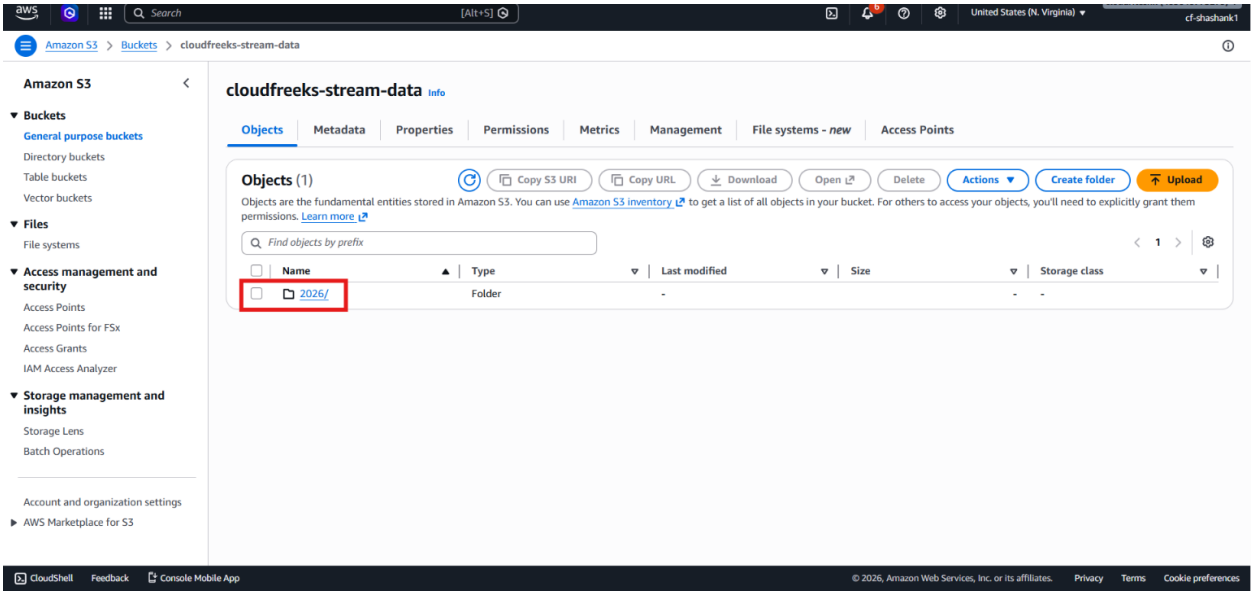
- Let Firehose send data for a few seconds.
- Click **Stop sending demo data**.



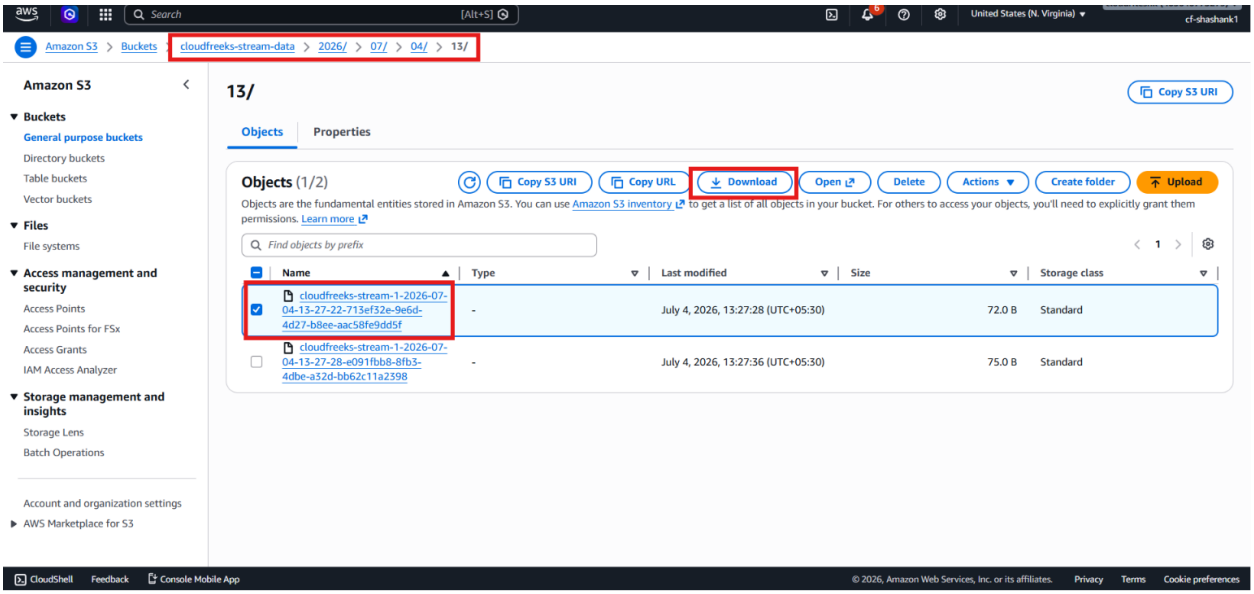
- This simulates a producer sending records into the Firehose stream.

Step 7 – Verify the Data in S3

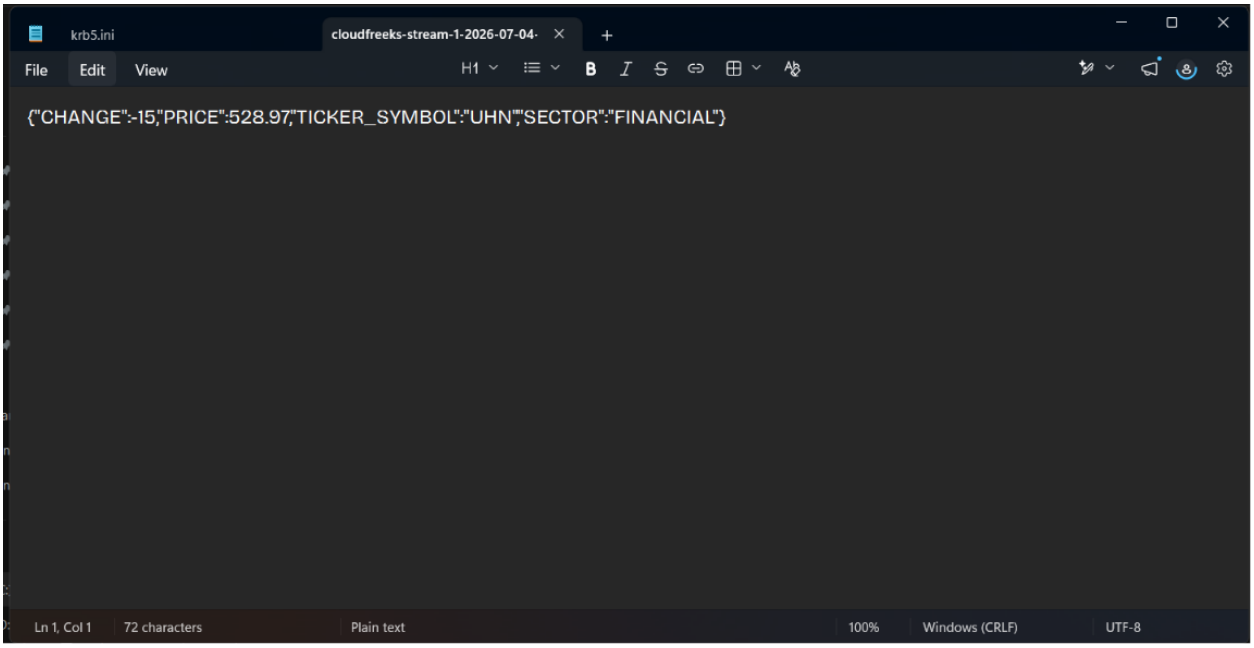
- Open the S3 bucket used as the destination.
- Refresh the bucket.
- Look for a generated folder prefix, such as a timestamp-based folder.



- Keep opening folders if there are multiple ones till you find file.
- Open the created file.



- Download it and open it in **Notepad** or another text editor.



- Confirm that the file contains the streamed data, such as ticker values or sample records.

Verification

- An **Amazon Data Firehose** stream was created successfully.
- The source was set to **Direct PUT**.
- The destination was configured as **Amazon S3**.
- Buffer size and buffer interval were configured correctly.
- Demo data was sent through Firehose.
- Records appeared in the S3 bucket.
- The downloaded file contained the expected streamed data.